Distrusting the Out-Party: Partisan Disbelief and Biased Information Processing

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  - Populist Radical Right Parties, rising affective polarization
  - Fake news, disinformation, and contested information everywhere

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  - **Out-groups**: People who support opposing parties
  - Widespread **disbelief** about others' views and intentions
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- Today's focus
    - **Disbelief in out-group knowledge**
    - **In-group bias in information processing**

What we do and find

Survey experiments in the US and South Korea

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   - People put more weight on judgments of in-groups
3. Correcting disbelief reduces and often completely eliminates **in-group bias**
4. Correcting disbelief reduces affective polarization, albeit inconclusive

Today's Plan

Surveys, Background

Study 1. Baseline Evidence of Disbelief

- Survey Design and Hypotheses
- Existence of Disbelief on Out-group's Knowledge

Study 2. Correcting Disbelief

- Survey Design and Hypotheses
- H1: Treatment Effects on Disbelief
- H2: Existence of In-group Bias in Information Processing
- H3: Treatment Effects of Correcting Disbelief on In-group Bias
- H4: Treatment Effects of Correcting Disbelief on Polarization

Conclusion

Surveys

- US and South Korea
- Two surveys
  - Study 1 (N$\approx$1,500/country): Document disbelief in out-group knowledge
  - Study 2 (N$\approx$4,200/country):
    - Document in-group bias in belief updating
    - Test whether correcting disbelief reduces the in-group bias
- Recruit participants through an online survey panel provider

Background/Acronyms/Abbreviations

- RP: Right-wing parties
  - US: Republican Party–current majority + president
  - SK: **People's Power Party (PPP)**
- LP: Left-wing parties
  - US: Democratic Party
  - SK: **Democratic Party of Korea (DPK)**–current majority + president
- NP: Non-partisans
- Drop others in SK

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Study 1: Survey Structure (N≈1,500/country)

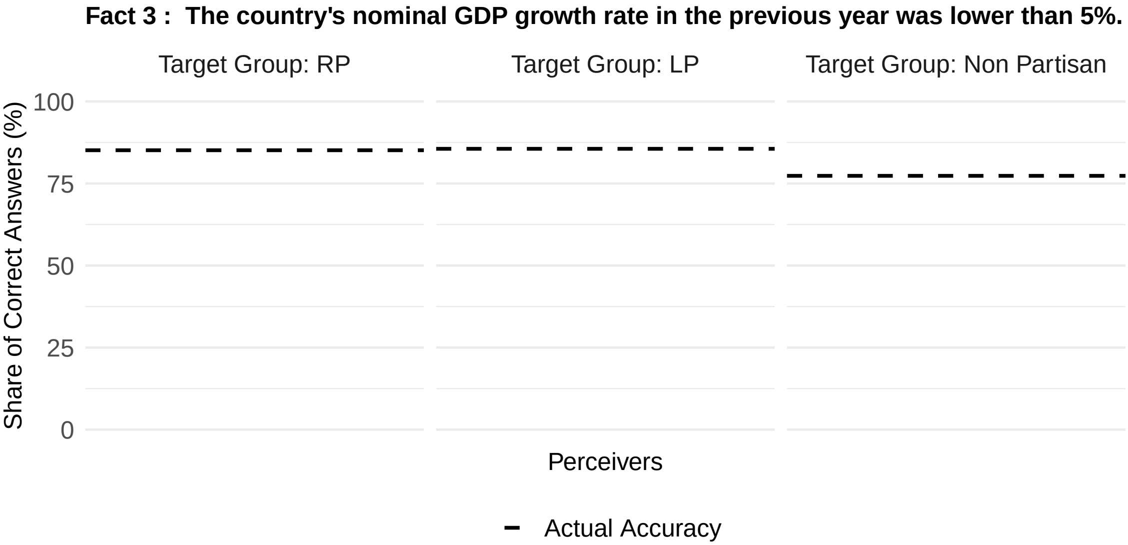
- Ask to evaluate (T or F) 8 factual questions:
- Examples
  - ”New Zealand is located in the Middle East.”
  - ”The country's GDP growth rate in the previous year was lower than 5%.”
- Ask to give confidence level
- Then, for each question, ask to estimate the accuracy rates for three groups
  - $p^t_{i,j}$: individual i's estimate on group t's accuracy rate on task/question j
  - $t \in \{RP, LP, NP\}, j = 1, ..., 8$
- A follow-up RCT with monetary incentives for the US

▸ Sum. Stat.

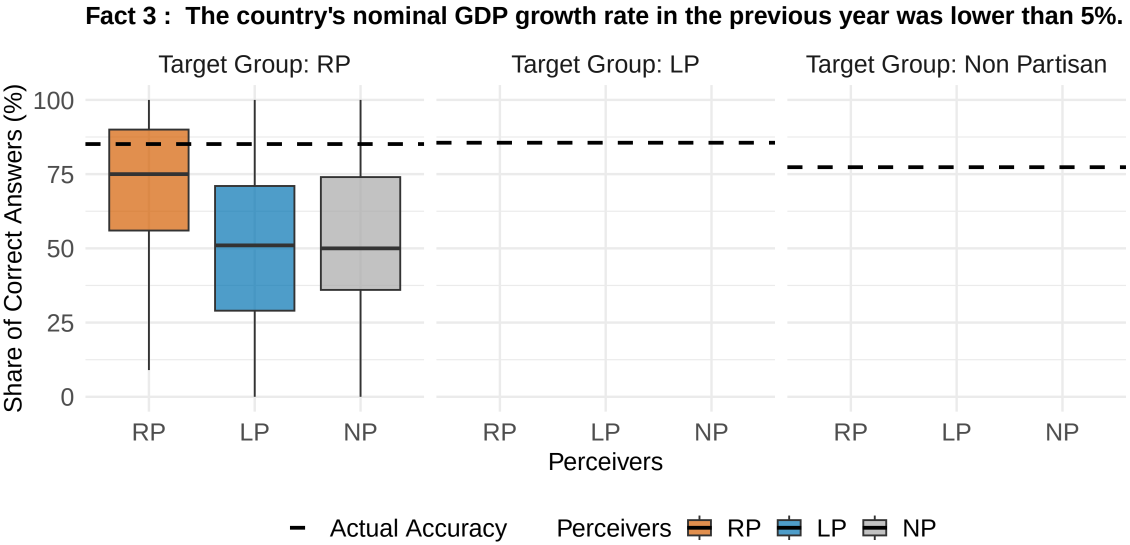
Today's Plan

SK Fact 3: GDP growth rate is less than 5%

Sum. Stat.

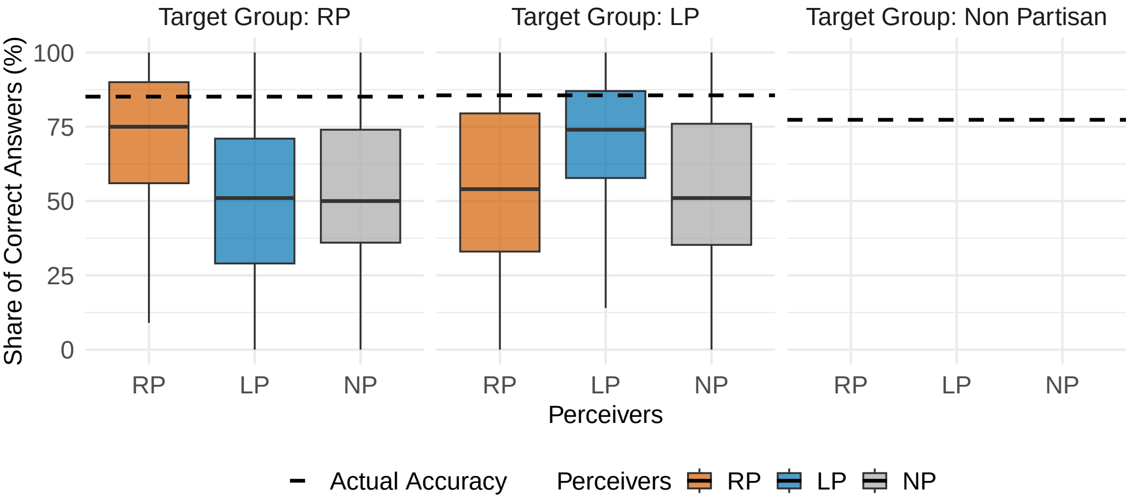


SK Fact 3: GDP growth rate is less than 5% ▸ Sum. Stat.



SK Fact 3: GDP growth rate is less than 5% ▸ Sum. Stat.

Fact 3 : The country's nominal GDP growth rate in the previous year was lower than 5%.



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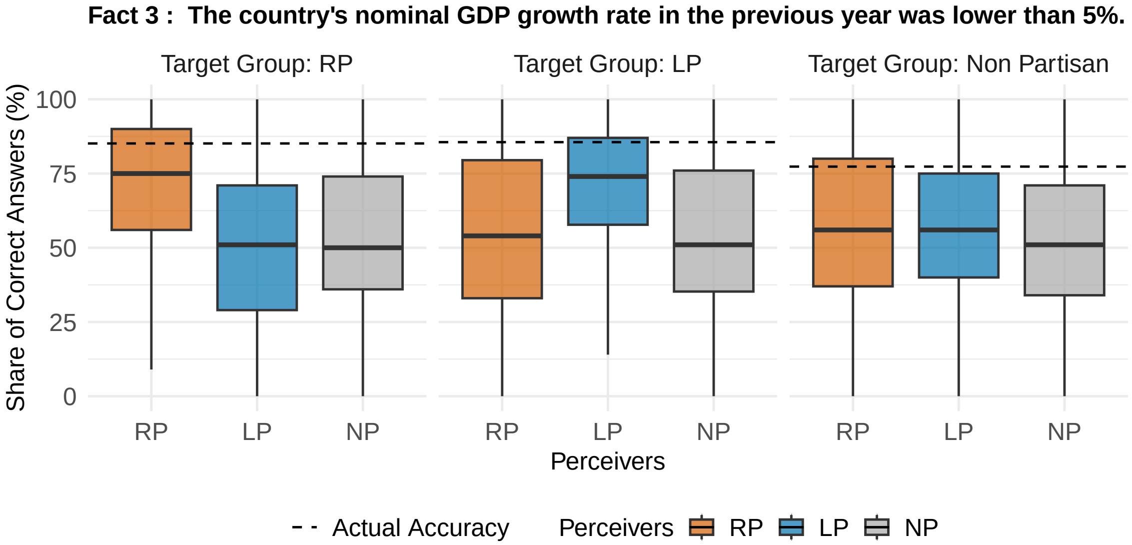
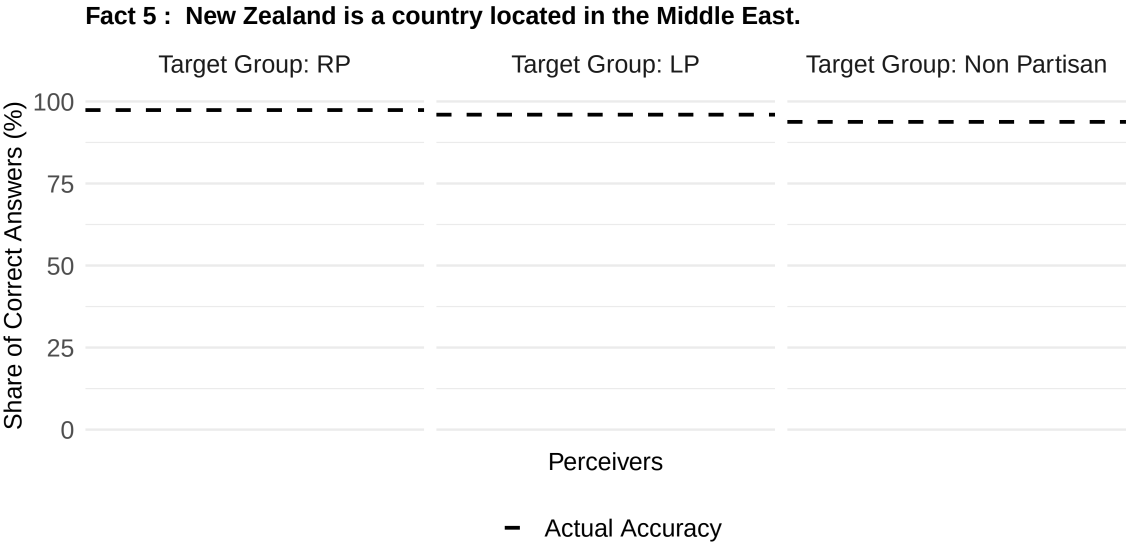


SK Fact 5: New Zealand is in the Middle East? ▸ Sum. Stat.



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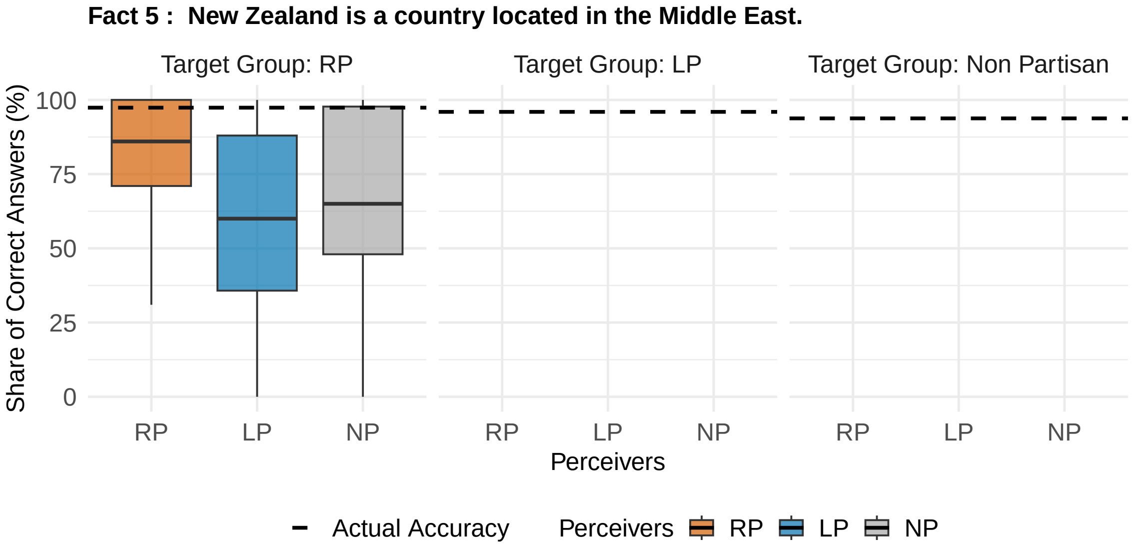

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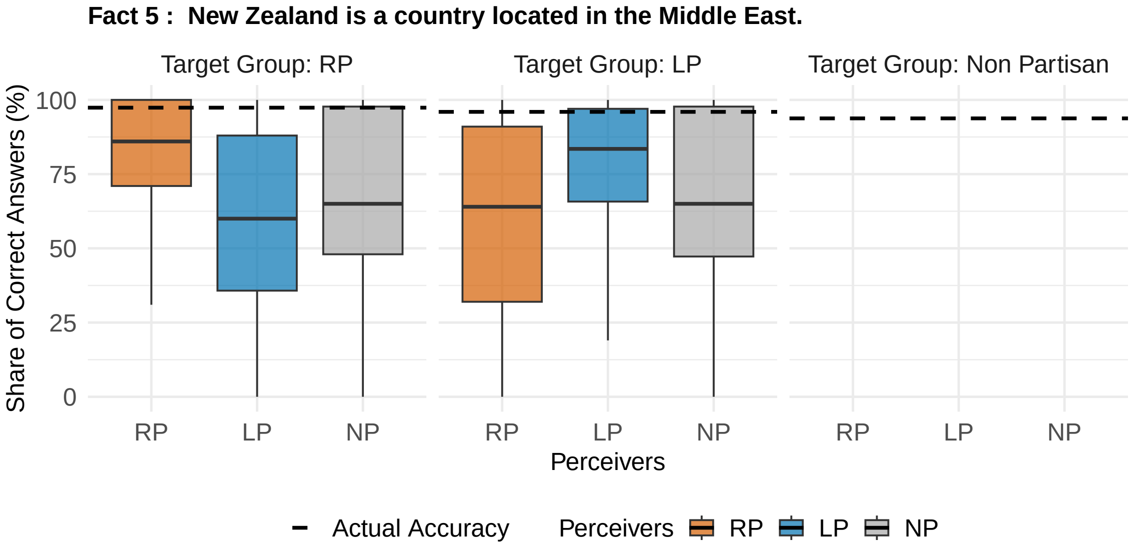

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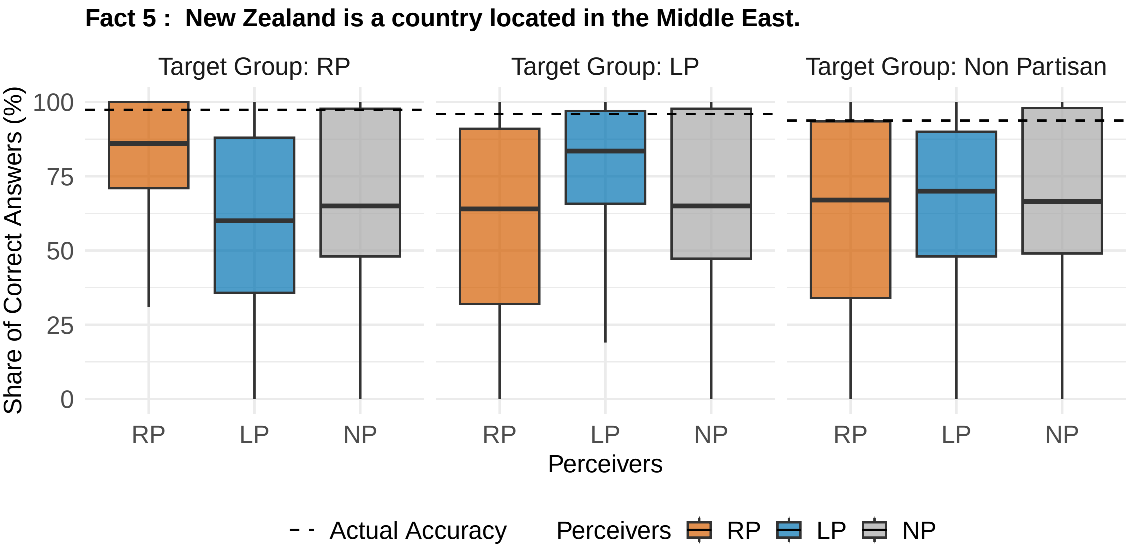

More Systematic Approach for Average Disbelief ▸ Sum. Stat.

- $p^t_{i,j}$: Accuracy rate of group $t \in \{RP, LP\}$ in task j perceived by i, $g(i)$ i's group

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- Target-based Partisan Disbelief (given perceiver i)

$$p^t_{i,j} = \beta_1 \mathbb{1}\{t = RP\} + \eta_i + \eta_j + \varepsilon^t_{i,j}$$

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- Null for Non-partisan (given perceiver i, $g(i) =$NP)

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$$p^t_{i,j} = \beta_2 \mathbb{1}\{t = RP\} + \eta_i + \eta_j + \varepsilon^t_{i,j}$$

  - Expect $\beta_2 = 0$

Average Disbelief is about 16-18 points ▸ Sum. Stat.

Country	SK	SK	SK	US	US	US
Perceiver	RP	LP	NP	RP	LP	NP
	(1)	(2)	(3)	(4)	(5)	(6)
Target = RP						
Target = LP						
Observations	5488	9536	7200	9872	9536	5680

- Note: targets are RP or LP; by fixed perceiver groups, individual + task FEs

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	(0.014)		(0.007)	(0.009)		(0.012)
Target = LP		0.172			0.166	
		(0.009)			(0.010)	
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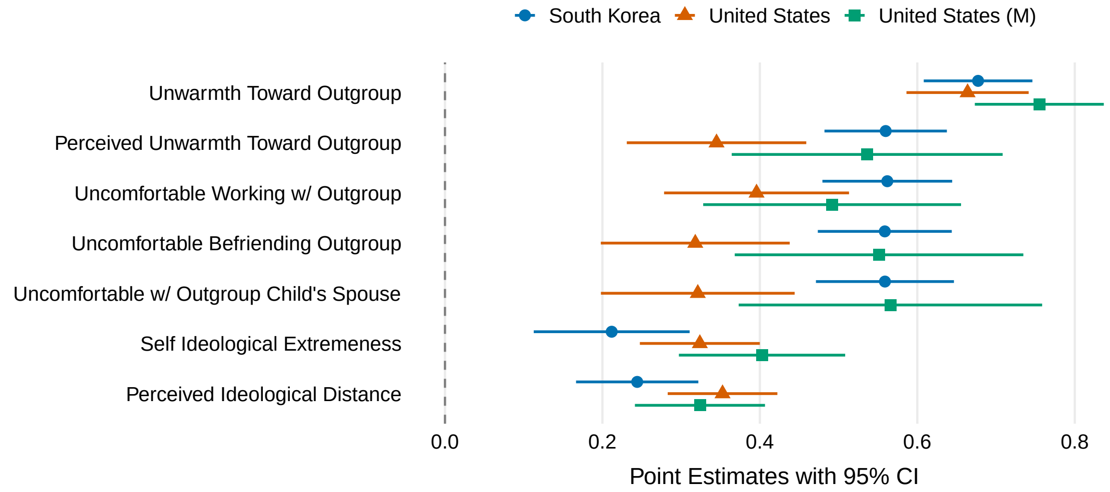
Disbelief Correlates with Ideological/Affective Polarization



Summary of Study 1

- In fact, both partisans are equally knowledgeable
- However, there are about 16–18 points of disbelief in out-group knowledge
- Non-partisans equally perceive knowledge of RP and LP
- Correlates with ideological and affective polarization

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Goals of Study 2

Given the baseline results in Study 1, we want to

- Document **in-group bias in information processing**
  - e.g., RP overweighs the opinion of RP over that of LP

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Given the baseline results in Study 1, we want to

- Document **in-group bias in information processing**
  - e.g., RP overweighs the opinion of RP over that of LP
- Run experiments if correcting **disbelief** reduces the **in-group bias**
  - Study 1 already shows RP and LP are, in fact, equally knowledgeable
  - Treatment = telling the fact above

Study 2: Survey Structure

1. Demographic questions

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4. Post-treatment judgment questions with **signals** *(details: next slide)*
   (Random at indiv. level within each question)
   - In-group signal: tells that in-groups know the correct answers
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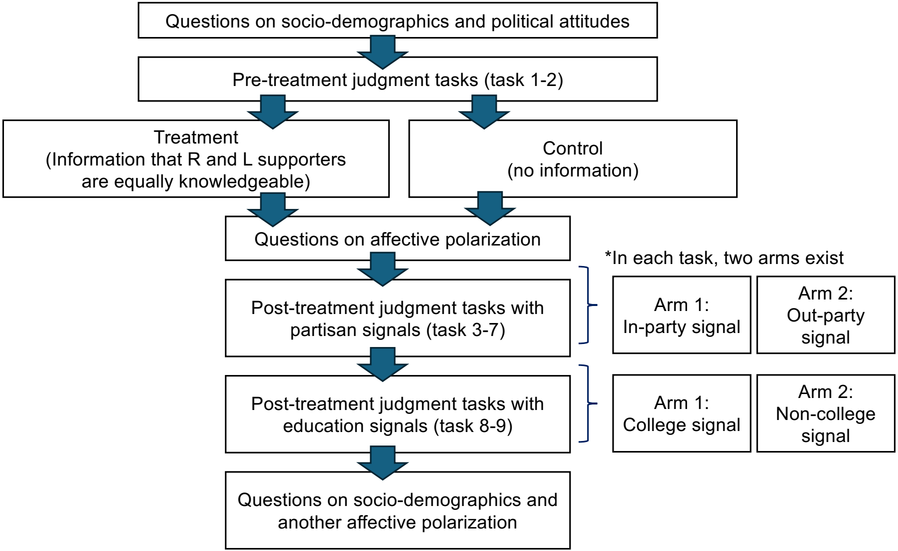
Only keep RP/LP with pre-treatment disbelief $\geq$ 5 pt (2314 in SK, 2879 in US)


Questions on socio-demographics and political attitudes
Pre-treatment judgment tasks (task 1-2)
Treatment
(Information that R and L supporters are equally knowledgeable)
Control
(no information)
Questions on affective polarization
*In each task, two arms exist
Post-treatment judgment tasks with partisan signals (task 3-7)
Arm 1:
In-party signal
Arm 2:
Out-party signal
Post-treatment judgment tasks with education signals (task 8-9)
Arm 1:
College signal
Arm 2:
Non-college signal
Questions on socio-demographics and another affective polarization

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What we want: See whether they update their beliefs (judgment & confidence)

Today's Plan

H1: Treatment Reduces Disbelief

- Define average out-group disbelief for post-treatment facts $\text{disbelief}_i^{\text{post}}$
- We run

$$\text{disbelief}_i^{\text{post}} = \alpha T_i + \varepsilon_i$$

	(1) SK	(2) US
Treatment		
Observations	2314	2879
Mean of outcome	0.231	0.196

▸ Conspiracy ▸ Sum. Stat.

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	(0.010)	(0.009)
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Conspiracy Sum. Stat.

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Measurement of In-group Bias in Information Processing

For individual i and task j, construct two types of dummy variables

1. Correct Judgment: $y_{i,j}^{J} \equiv \mathbb{1}\{ J_{i,j}^{1} - J_{i,j}^{0} > 0\}$;

Measurement of In-group Bias in Information Processing

For individual *i* and task *j*, construct two types of dummy variables

1. Correct Judgment: $y^J_{i,j} \equiv \mathbb{1}\{ J^1_{i,j} - J^0_{i,j} > 0\}$;
   - $J^0_{i,j}$: Correctness before signals ($J^0_{i,j} = 1$ if Correct and $= 0$ if Wrong)
   - $J^1_{i,j}$: Correctness after signals

Measurement of In-group Bias in Information Processing

For individual i and task j, construct two types of dummy variables

1. Correct Judgment: $y^J_{i,j} \equiv \mathbb{1}\{ J^1_{i,j} - J^0_{i,j} > 0\}$;

2. Confidence towards Correct Answer: $y^\mu_{i,j} \equiv \mathbb{1}\{\mu^1_{i,j} - \mu^0_{i,j} > 0\}$;
   - $\mu^0_{i,j}$: Confidence towards Correct answers before signals

$$\mu^0_{i,j} = \begin{cases} \frac{a^0_{i,j}}{100} & \text{if } J^0_{i,j} = 1 \\ 1 - \frac{a^0_{i,j}}{100} & \text{if } J^0_{i,j} = 0 \end{cases}$$

   where $a^0_{i,j} \in [0, 100]$ is confidence level for their answer
   - $\mu^1_{i,j}$: Confidence towards Correct answers after signals

H2: In-group Signals Shift Beliefs More Toward the Truth?

Specification: (i: indiv., j: task)

$$y_{i,j} = \beta \mathbb{1}\{\text{In-group Signal}\}_{i,j} + \eta_j + \varepsilon_{i,j}$$

- $y_{i,j}$: measure of information processing, $y^J_{i,j}$ or $y^\mu_{i,j}$
- $\mathbb{1}\{\text{In-group Signal}\}_{i,j}$: dummy if in-group signal
  - e.g.) If R, "The majority of R says this is True..." is an in-group signal
- Expect $\beta > 0$

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	(1) SK Dummy	(2) SK Continuous	(3) US Dummy	(4) US Continuous
In-Group Signal				
Observations	3432	3432	4353	4353
Mean of outcome	0.103	0.425	0.170	0.480

▸ Education Signal ▸ Sum. Stat.

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	(0.011)	(0.017)	(0.012)	(0.015)
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  - e.g.) ”You think that R are more knowledgeable than D. This is wrong.”

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  - e.g.) ”You think that R are more knowledgeable than D. This is wrong.”
- See if treatment reduces in-group bias in information processing

H3: Treatment, In-group Bias in Information Processing

Specification ($s_{i,j} = I$: In-group signal)

$$y_{i,j} = \beta_1 \mathbb{1}\{s_{i,j} = I\} + \beta_2 T_i + \beta_3 \left(\mathbb{1}\{s_{i,j} = I\} \times T_i\right) + \eta_j + \varepsilon_{i,j}$$

- Expect $\beta_3 < 0$ (given that $\beta_1 > 0$)

▸ Sum. Stat.

H3: Treatment, In-group Bias in Information Processing

$$y_{i,j} = \beta_1 \mathbb{1}\{s_{i,j} = I\} + \beta_2 T_i + \beta_3 \left(\mathbb{1}\{s_{i,j} = I\} \times T_i\right) + \eta_j + \varepsilon_{i,j}$$

	(1) SK Dummy	(2) SK Continuous	(3) US Dummy	(4) US Continuous
In-Group Signal				
Treatment				
In-Group Signal x Treatment				
Observations	6942	6942	8637	8637
Mean of outcome	0.103	0.421	0.164	0.481

▸ Sum. Stat.

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	(1) SK Dummy	(2) SK Continuous	(3) US Dummy	(4) US Continuous
In-Group Signal	0.057			
	(0.011)			
Treatment	0.016			
	(0.009)			
In-Group Signal x Treatment	-0.031			
	(0.002)			
Observations	6942	6942	8637	8637
Mean of outcome	0.103	0.421	0.164	0.481

▸ Sum. Stat.

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	(1)	(2)	(3)	(4)
	SK	SK	US	US
	Dummy	Continuous	Dummy	Continuous
In-Group Signal	0.057	0.076		
	(0.011)	(0.008)		
Treatment	0.016	0.021		
	(0.009)	(0.018)		
In-Group Signal x Treatment	-0.031	-0.058		
	(0.002)	(0.026)		
Observations	6942	6942	8637	8637
Mean of outcome	0.103	0.421	0.164	0.481

Sum. Stat.

H3: Treatment, In-group Bias in Information Processing

$$y_{i,j} = \beta_1 \mathbb{1}\{s_{i,j} = I\} + \beta_2 T_i + \beta_3 \left(\mathbb{1}\{s_{i,j} = I\} \times T_i\right) + \eta_j + \varepsilon_{i,j}$$

	(1) SK Dummy	(2) SK Continuous	(3) US Dummy	(4) US Continuous
In-Group Signal	0.057	0.076	0.006	0.045
	(0.011)	(0.008)	(0.009)	(0.012)
Treatment	0.016	0.021	-0.019	0.020
	(0.009)	(0.018)	(0.008)	(0.018)
In-Group Signal x Treatment	-0.031	-0.058	0.015	-0.035
	(0.002)	(0.026)	(0.005)	(0.011)
Observations	6942	6942	8637	8637
Mean of outcome	0.103	0.421	0.164	0.481

▸ Sum. Stat.

Today's Plan

H4: Effects of Correcting Disbelief on Polarization

Specification (LHS: Combined measures of affective polarization)

$$\text{pol}_i = \gamma T_i + \varepsilon_i$$

- **Unfav**orable measure (unwarmth measure)
- **Uncomf**ortable with out-groups' colleagues/friends/spouses of children...

H4: Effects of Correcting Disbelief on Polarization

Specification (LHS: Combined measures of affective polarization)

$$\text{pol}_i = \gamma T_i + \varepsilon_i$$

- **Unfav**orable measure (unwarmth measure)
- **Uncomf**ortable with out-groups' colleagues/friends/spouses of children...

	(1) SK Unfav	(2) SK Uncomf	(3) US Unfav	(4) US Uncomf
Treatment	-0.024	-0.002	-0.078	-0.022
	(0.012)	(0.012)	(0.013)	(0.016)
Observations	2314	2314	2879	2879
Mean of outcome	0.527	0.433	0.490	0.268

Today's Plan

Conclusion

- **Widespread disbelief about out-group knowledge** for factual questions
- **In-group bias in information processing**
- Correcting the disbelief can reduce the in-group bias

Today's Plan

Summary Statistics

Hypotheses, Maths

Further Results

- Balanced Test
- Conspiracy Theory
- Education Signal

Appendix: Disbelief on All Facts

Study 1: Summary Statistics ▸ Back

	SK RP	SK LP	SK NP	US RP	US LP	US NP
Demographics						
Female Ratio	0.41	0.49	0.59	0.47	0.59	0.55
College-educated Ratio	0.83	0.77	0.75	0.52	0.54	0.37
Age (50+) Ratio	0.54	0.42	0.35	0.54	0.48	0.38
Judgments						
Average Accuracy Rate	0.74	0.74	0.73	0.67	0.63	0.57
Average Confidence	0.81	0.78	0.71	0.74	0.71	0.65
Observations	343	596	450	617	596	355

Study 2: Summary Statistics

Back

	SK Treated	SK Control	SK Diff	US Treated	US Control	US Diff
RP Supporters Ratio	0.256	0.249	0.007	0.543	0.522	0.021
			(0.018)			(0.019)
Female Ratio	0.482	0.484	-0.002	0.474	0.517	-0.043
			(0.021)			(0.019)
College-educated Ratio	0.804	0.795	0.010	0.547	0.531	0.016
			(0.017)			(0.019)
Age (50+) Ratio	0.709	0.709	-0.000	0.662	0.689	-0.027
			(0.019)			(0.017)
Pre-treatment Accuracy Rate	0.955	0.957	-0.002	0.849	0.832	0.017
			(0.006)			(0.009)
Pre-treatment Partisan Disbelief	0.351	0.342	0.009	0.342	0.337	0.005
			(0.011)			(0.009)
Observations	1170	1144		1428	1451	

Today's Plan

Summary Statistics

Hypotheses, Maths

Further Results

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Appendix: Disbelief on All Facts

Hypotheses: Disbelief in out-group knowledge

- Target-based Disbelief
  - RP supporters believe that RP supporters are more knowledgeable
  - LP supporters believe that LP supporters are more knowledgeable
  - Non-partisans believe that RP and LP supporters are equally knowledgeable

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  - RP supporters are seen as more knowledgeable by RP than LP
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  - NP supporters are seen as equally knowledgeable by LP and RP

Hypotheses: Disbelief in out-group knowledge

- Target-based Disbelief
  - RP supporters believe that RP supporters are more knowledgeable
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- Perceiver-based Disbelief
  - RP supporters are seen as more knowledgeable by RP than LP
  - LP supporters are seen as more knowledgeable by LP than RP
  - NP supporters are seen as equally knowledgeable by LP and RP
- Today: focus on Target-based Disbelief (both are almost identical)

Correlation with Affective Polarization

- Define "disbelief": simple difference between estimates for in-group and out-group

$$\text{disbelief}_{i,j} \equiv p_{i,j}^{g(i)} - p_{i,j}^{g(i)'}$$

$$\text{disbelief}_{i} \equiv \frac{1}{8}\sum_{j=1}^{8} \text{disbelief}_{i,j}$$

- Regress different polarization measures on disbelief_i
  - Polarization measures are standardized to [0,1]

Hypotheses:

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- H1: Treatment decreases post-treatment disbelief
    - Not just a manipulation check
    - Pre- and post-treatment questions are different

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- H2: Partisans have in-group bias in information processing

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- H1: Treatment decreases post-treatment disbelief
    - Not just a manipulation check
    - Pre- and post-treatment questions are different
- H2: Partisans have in-group bias in information processing
- H3: Treatment decreases the in-group bias in information processing

Hypotheses:

- H1: Treatment decreases post-treatment disbelief
  - Not just a manipulation check
  - Pre- and post-treatment questions are different
- H2: Partisans have in-group bias in information processing
- H3: Treatment decreases the in-group bias in information processing
- H4: Treatment decreases affective polarization

Today's Plan

Summary Statistics

Hypotheses, Maths

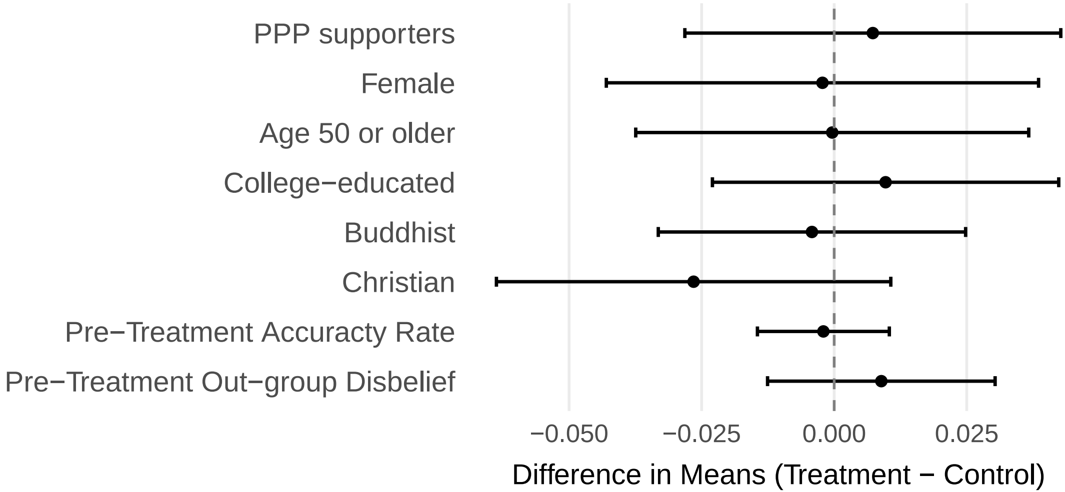
Further Results

- Balanced Test
- Conspiracy Theory
- Education Signal

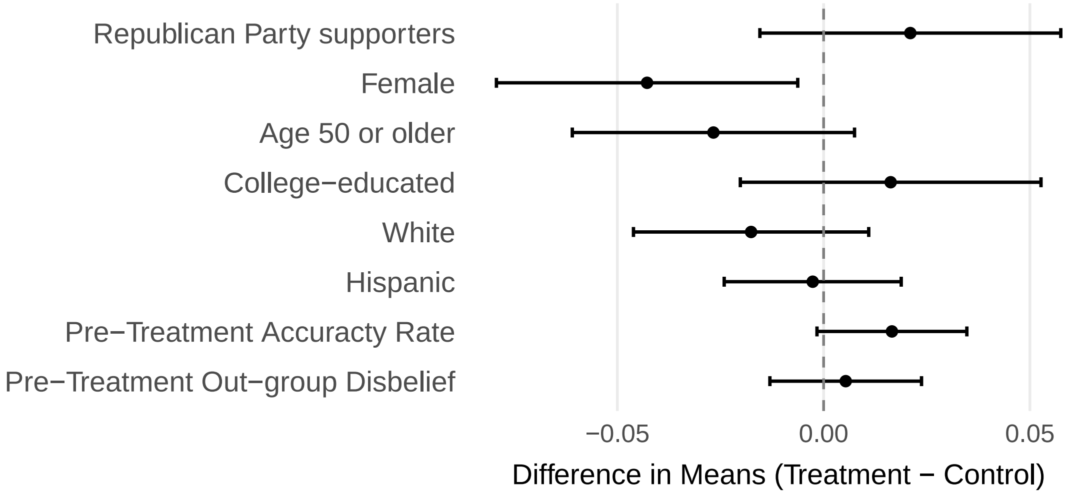
Appendix: Disbelief on All Facts

Today's Plan

H3: Balanced Test across Control and Treated



H3: Balanced Test across Control and Treated



Today's Plan

Summary Statistics

Hypotheses, Maths

Further Results

- Balanced Test
- Conspiracy Theory
- Education Signal

Appendix: Disbelief on All Facts

H1-A: Treatment Reduces Disbelief in Conspiracy Theory

Back

	(1) SK	(2) US
Treatment	-0.028	-0.063
	(0.015)	(0.013)
Observations	2314	2879
Mean of outcome	0.343	0.239

Today's Plan

Summary Statistics

Hypotheses, Maths

Further Results

- Balanced Test
- Conspiracy Theory
- Education Signal

Appendix: Disbelief on All Facts

H2: Partisan bias is larger than education-based bias

South Korea:

- Compare information processing based on college vs non-college signals
- Restrict samples to control group whose pre-signal answers were wrong

	(1) Dummy Educ	(2) Dummy Educ	(3) Dummy Party	(4) Cont. Educ	(5) Cont Educ	(6) Cont Party
In-Group Signal		-0.004	0.171		-0.025	0.140
		(0.014)	(0.028)		(0.024)	(0.029)
College Signal	0.006			-0.003		
	(0.014)			(0.025)		
Observations	1191	1191	1138	1191	1191	1138
Mean of outcome	0.070	0.070	0.310	0.257	0.257	0.529

▸ Back

H2: Partisan bias is larger than education-based bias

United States:

- Compare information processing based on college vs non-college signals
- Restrict samples to control group whose pre-signal answers were wrong

	(1) Dummy Educ	(2) Dummy Educ	(3) Dummy Party	(4) Cont Educ	(5) Cont Educ	(6) Cont Party
In-Group Signal		-0.042	0.023		-0.040	0.091
		(0.035)	(0.025)		(0.033)	(0.025)
College Signal	0.032			0.052		
	(0.035)			(0.033)		
Observations	847	847	1593	847	847	1593
Mean of outcome	0.521	0.521	0.464	0.603	0.603	0.530

Back

Today's Plan

Summary Statistics

Hypotheses, Maths

Further Results
- Balanced Test
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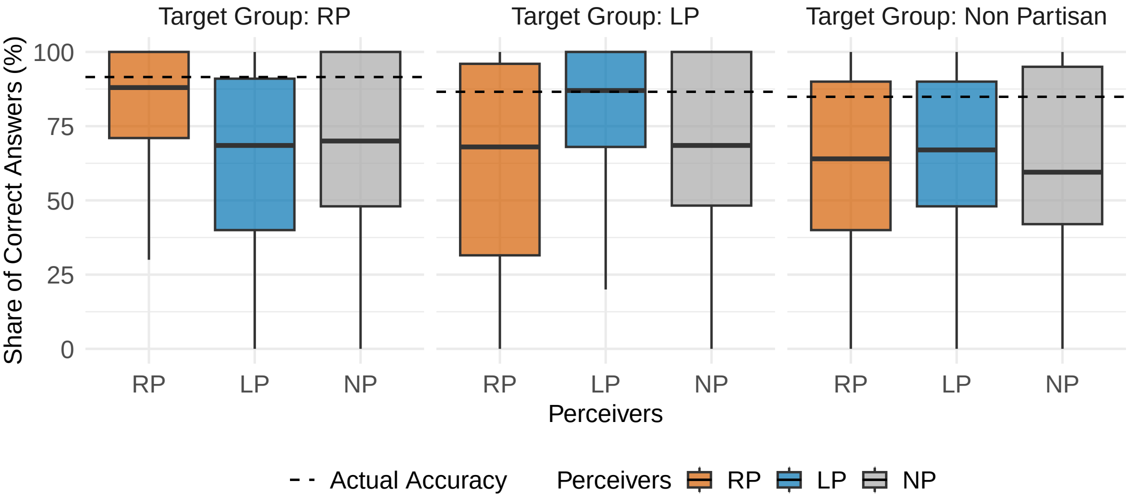
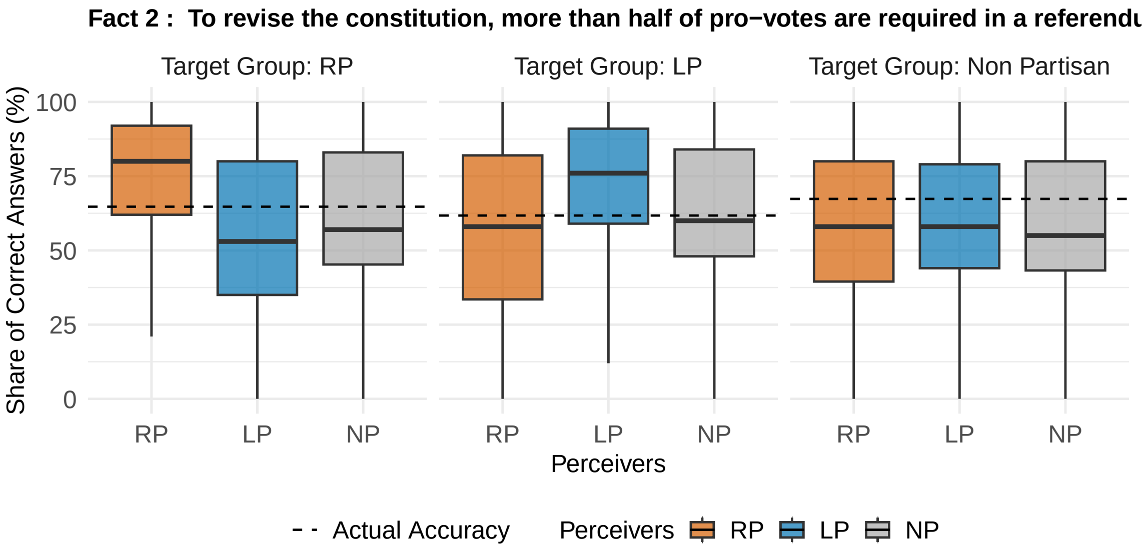
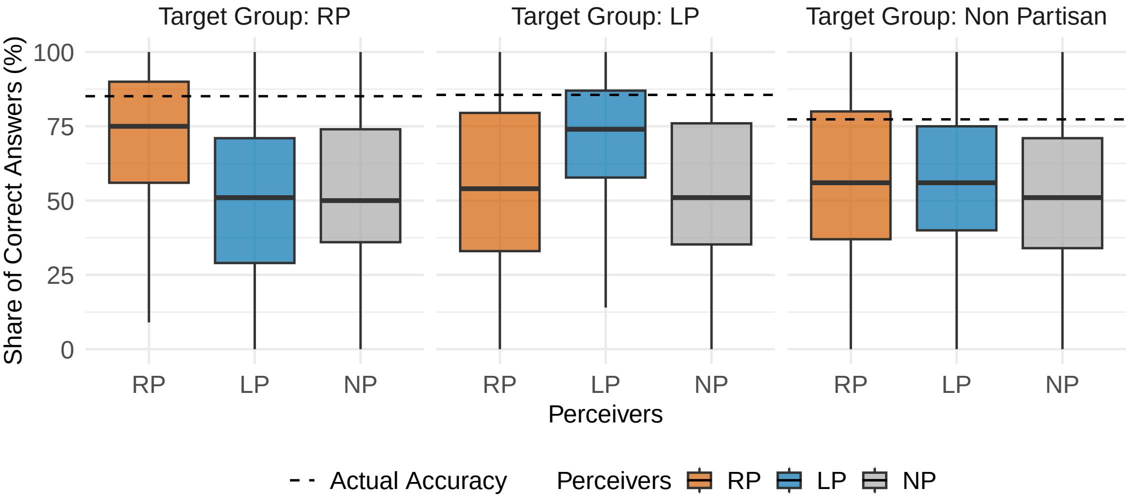
Appendix: Disbelief on All Facts

Fact 1 : The term of office of the National Assembly is 2 years.

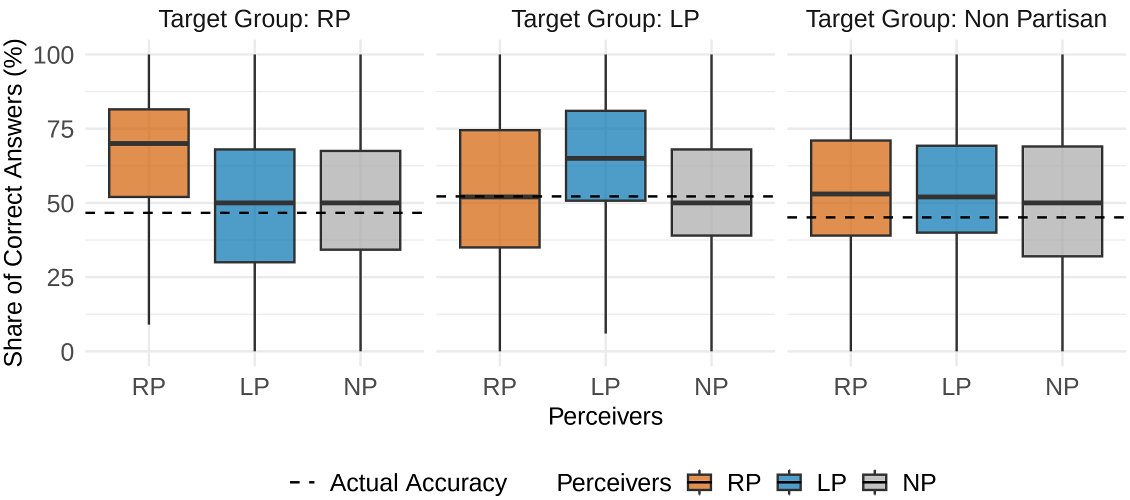



Fact 2 : To revise the constitution, more than half of pro−votes are required in a referendu
Target Group: RP
Target Group: LP
Target Group: Non Partisan
Share of Correct Answers (%)
100
75
50
25
0
RP
LP
NP
Perceivers
Actual Accuracy
Perceivers
RP
LP
NP

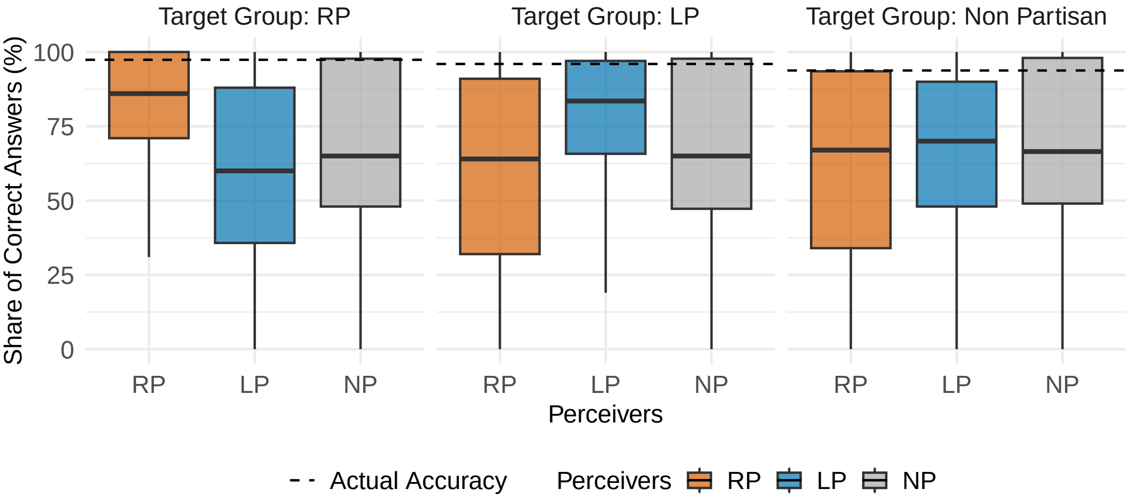
Fact 3 : The country's nominal GDP growth rate in the previous year was lower than 5%.



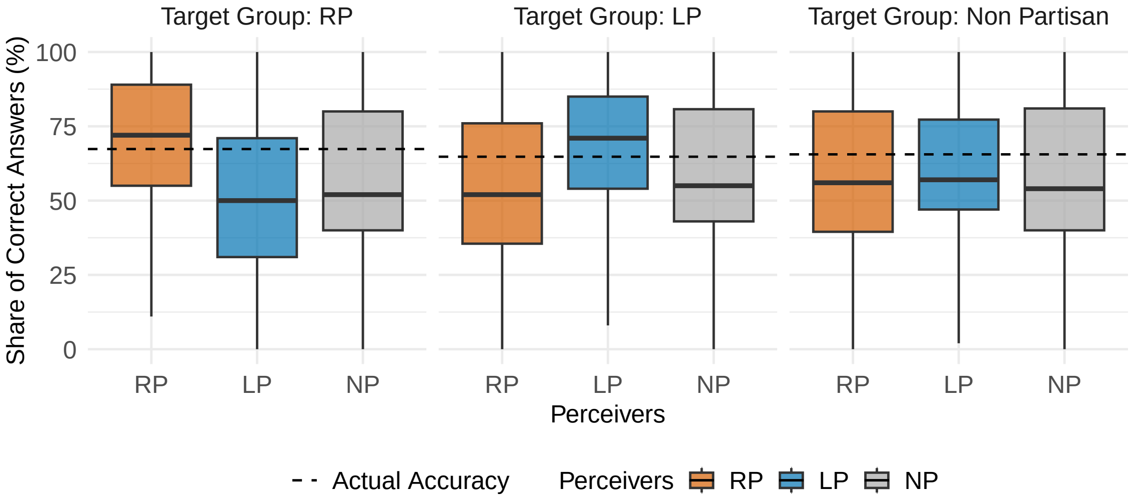
Fact 4 : For every hundred working−age people, there are forty old−age people who is 65



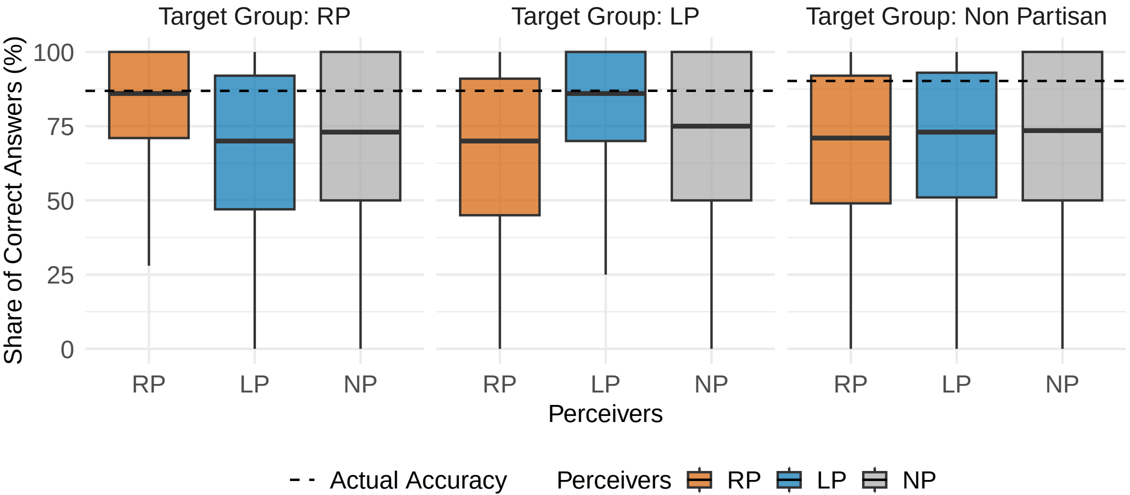
Fact 5 : New Zealand is a country located in the Middle East.



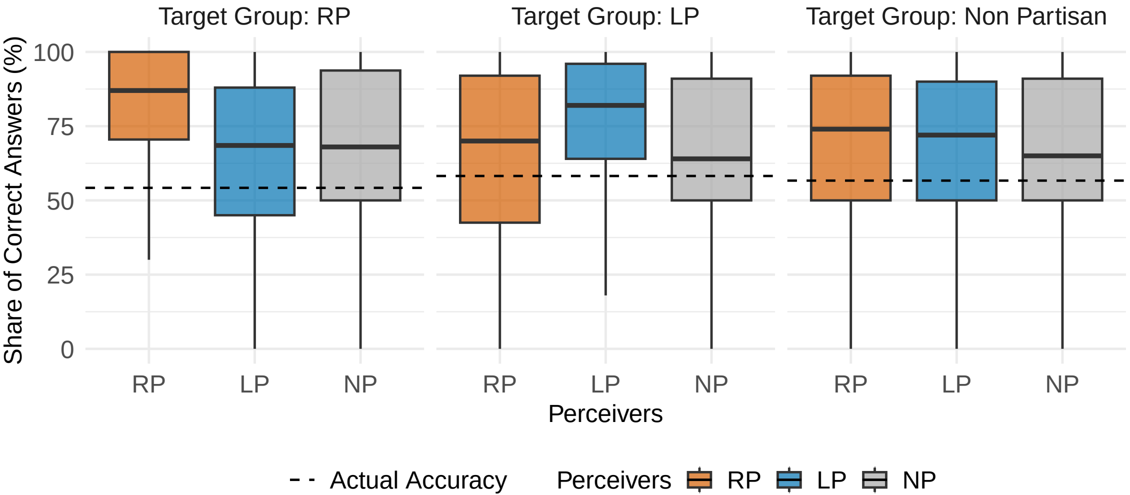
Fact 6 : iPhone was invented before 2000.



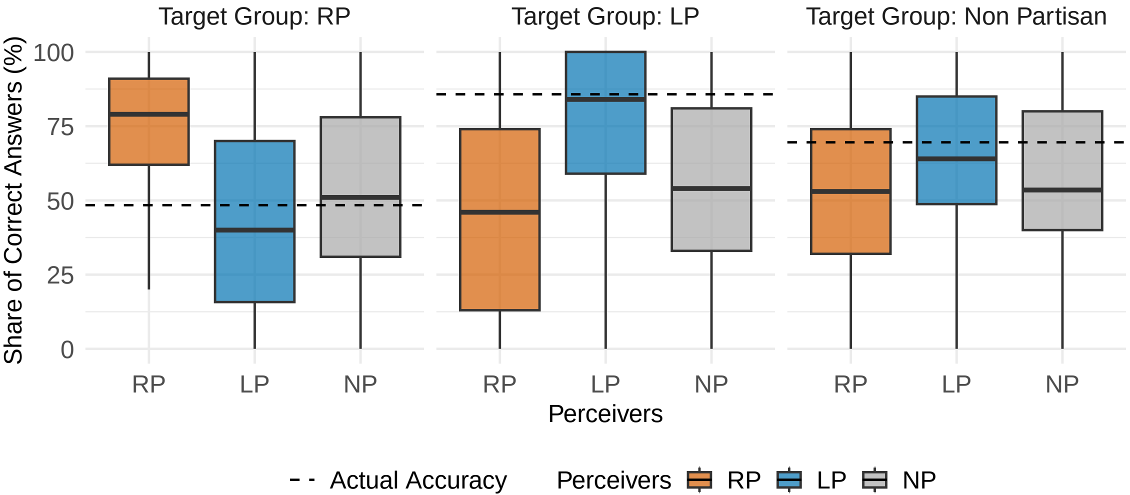
Fact 7 : It is stipulated by law that one must be at least 19 years old to drink alcohol.



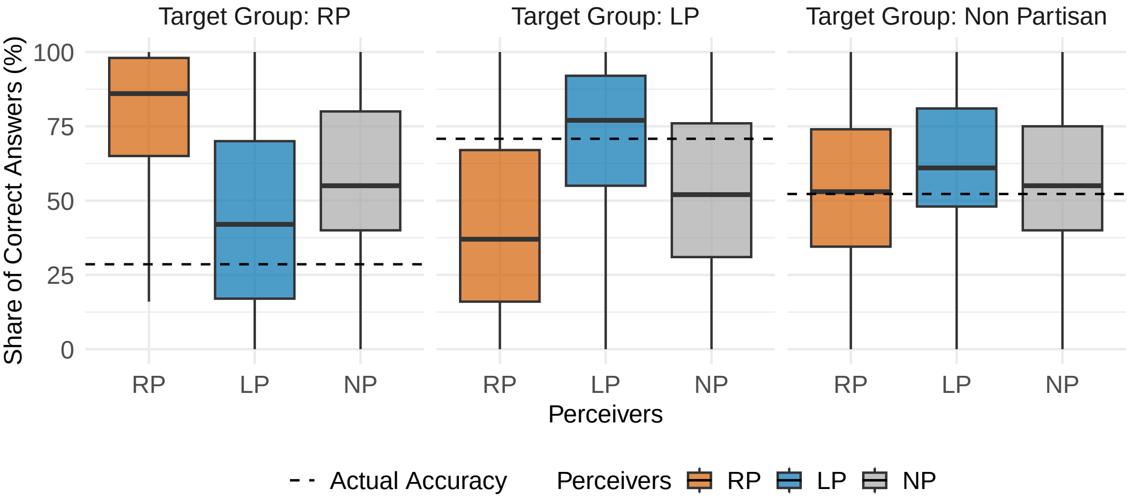
Fact 8 : The highest mountain in the country is Hallasan Mountain.



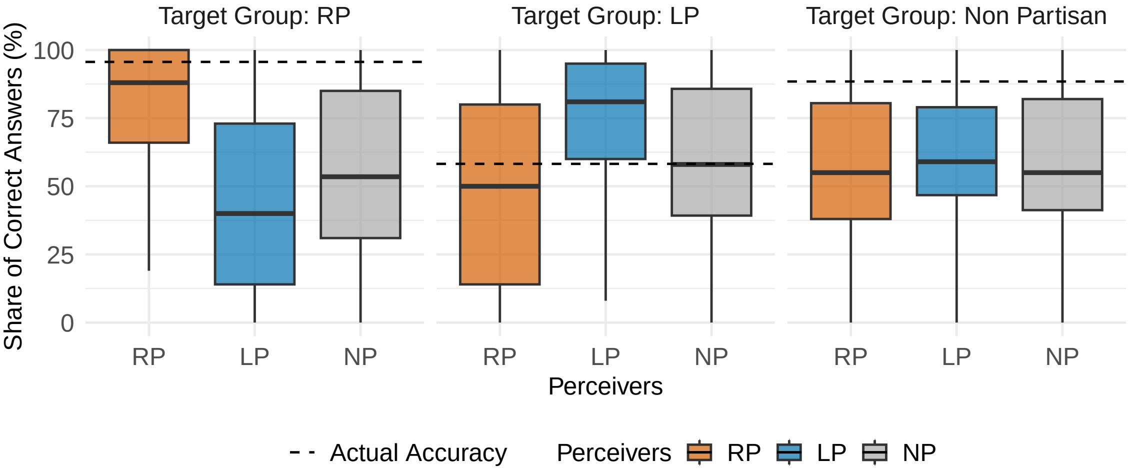
Fact 9 : There were widespread election fraud in the 2020 and 2024 elections.



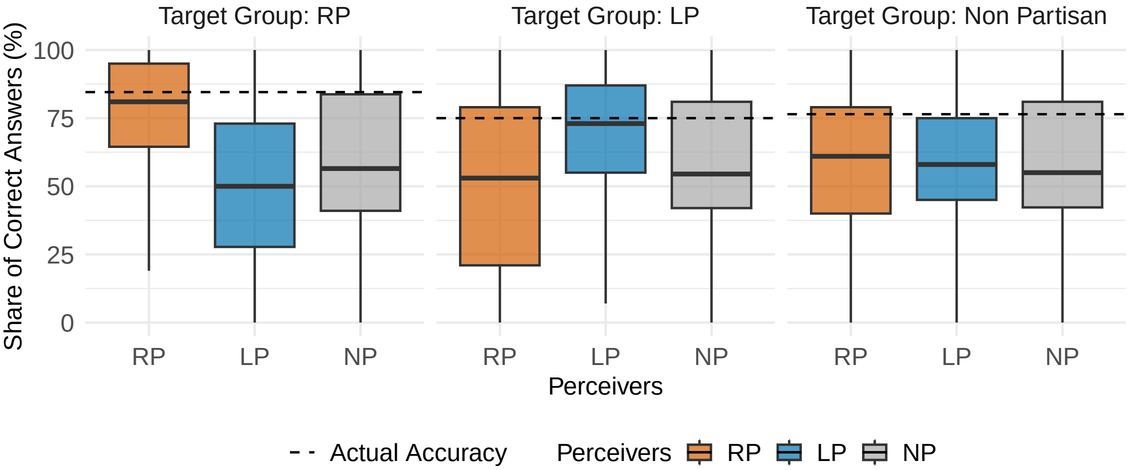
Fact 10 : China is infiltrating South Korean institutions, aiming to undermine the nation's



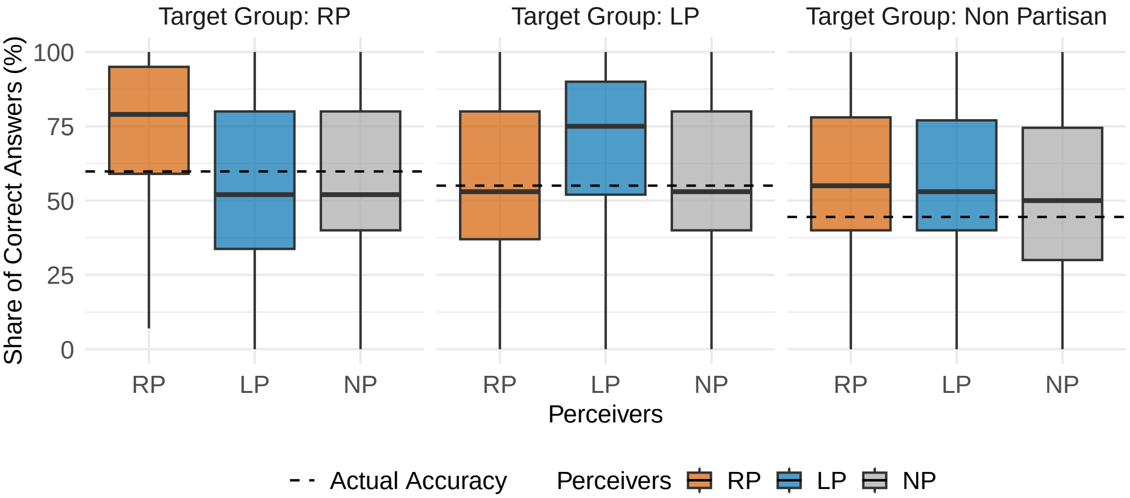
Fact 11 : The Supreme Court disqualified Lee from the presidential election in collaboration with Yoon Suk Yeol.

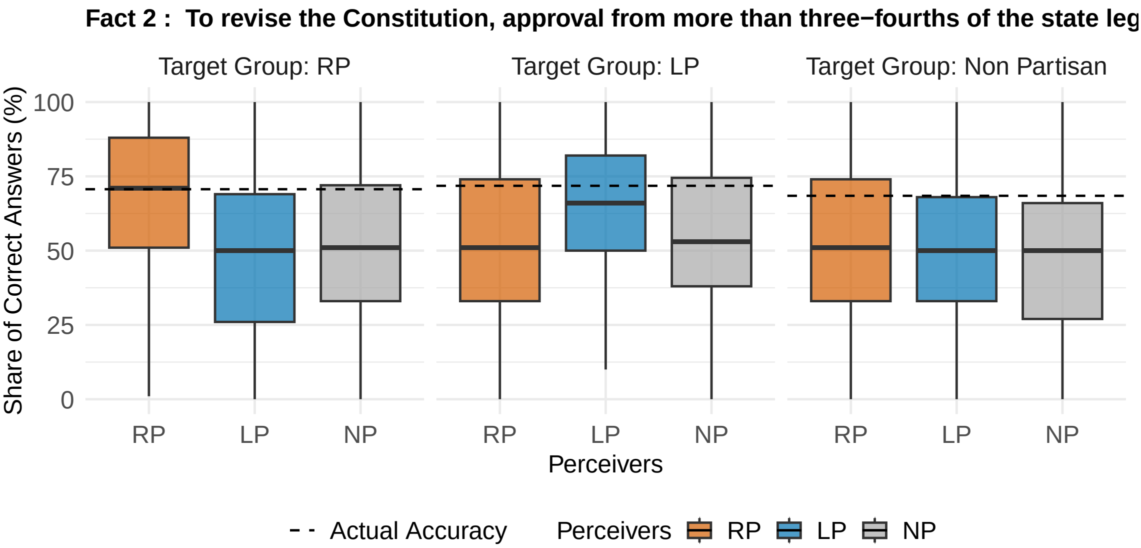


Fact 12 : U.S. government controls major political decisions in South Korea, including opposition crackdowns.



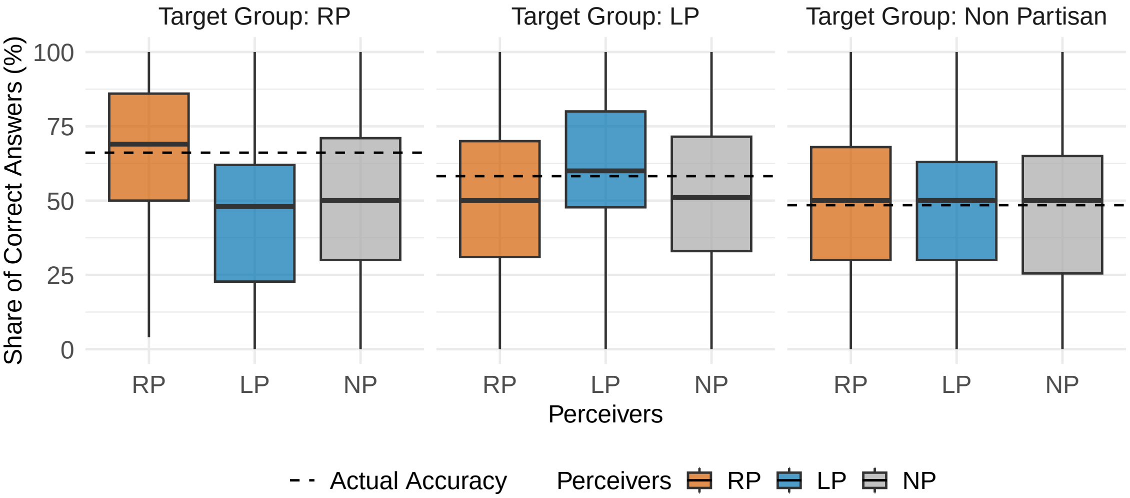
Fact 1 : The term of office of the Senate is 4 years.



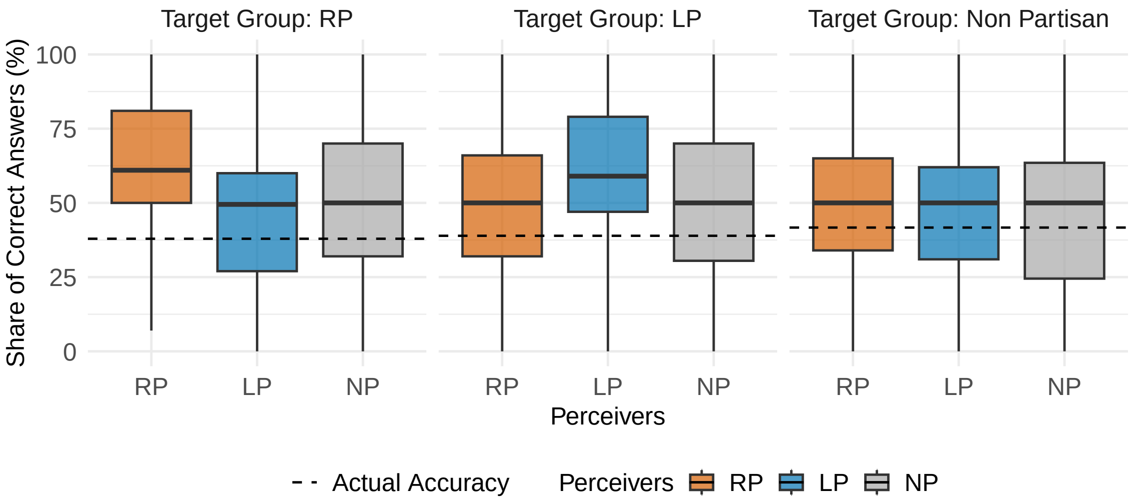


Fact 2 : To revise the Constitution, approval from more than three−fourths of the state leg
Target Group: RP
Target Group: LP
Target Group: Non Partisan
Share of Correct Answers (%)
100
75
50
25
0
RP
LP
NP
RP
LP
NP
RP
LP
NP
Perceivers
Actual Accuracy
Perceivers
RP
LP
NP

Fact 3 : The country's nominal GDP growth rate in the previous year was lower than 7%.



Fact 4 : For every hundred working−age people, there are forty old−age people who are 6



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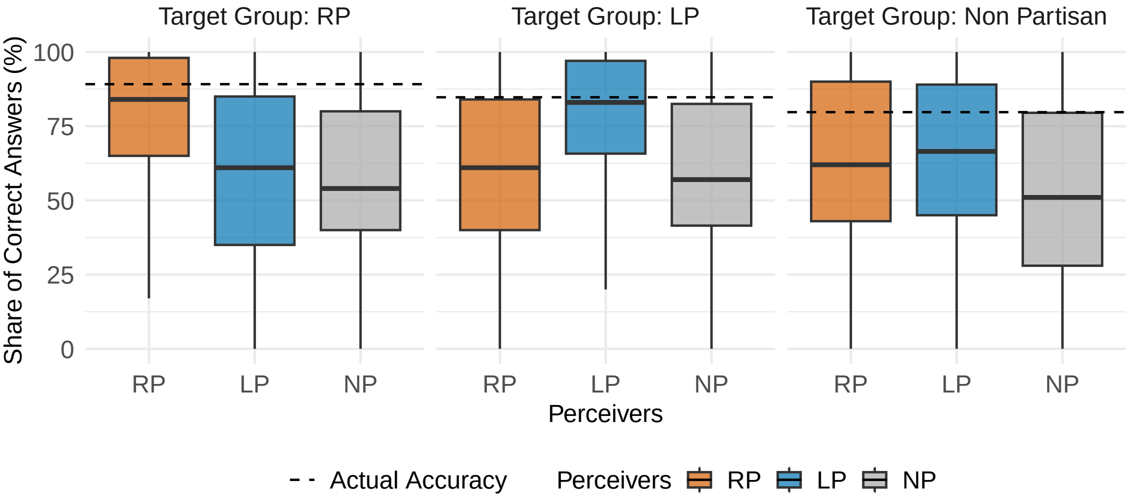
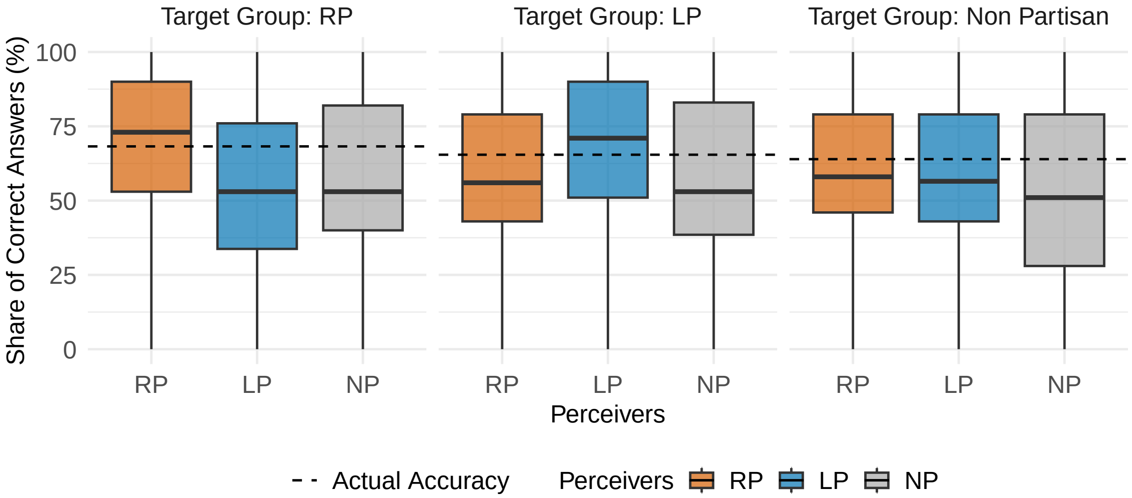
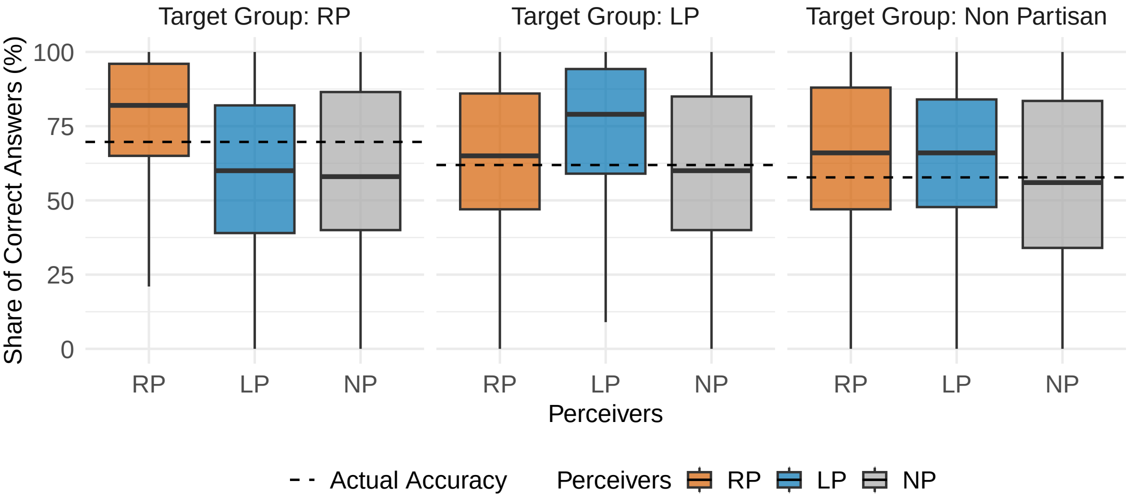
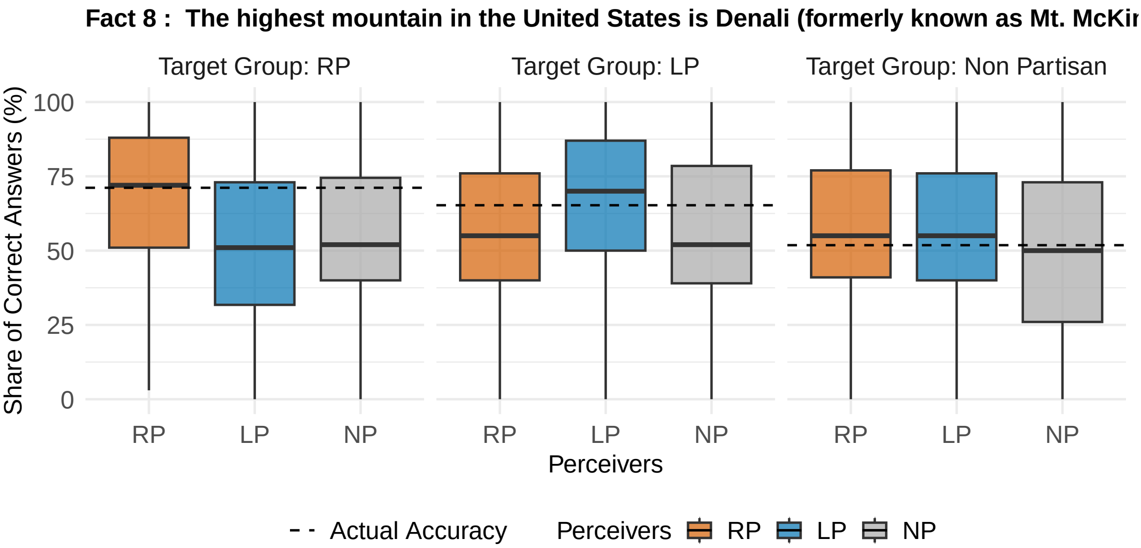


Fact 6 : iPhone was invented before 2000.



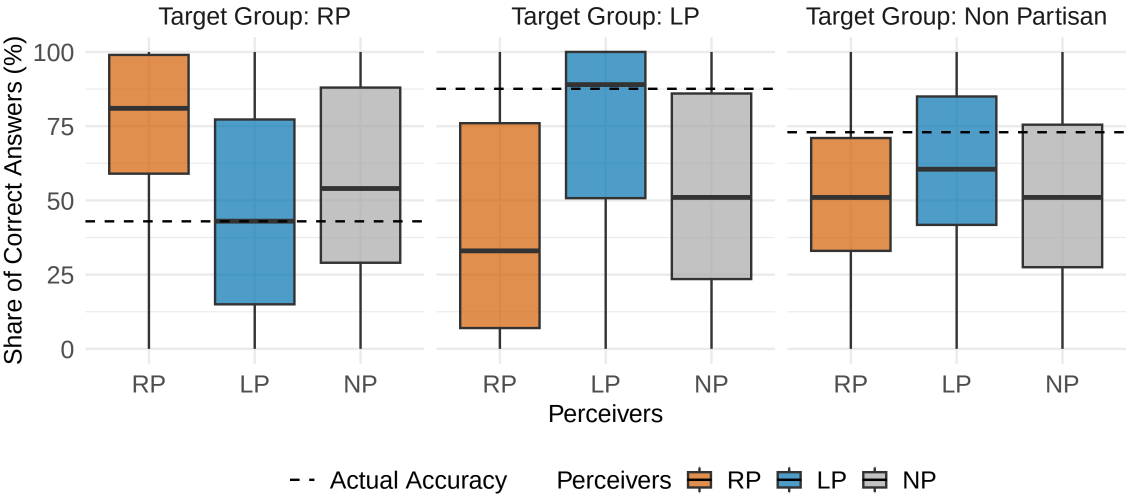
Fact 7 : The largest state in the United States is Alaska.

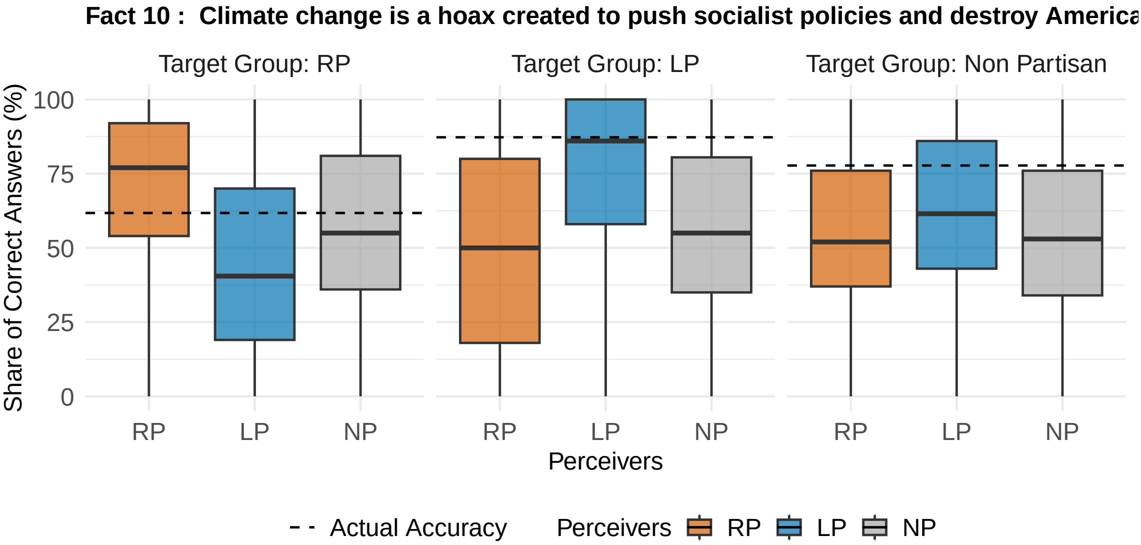




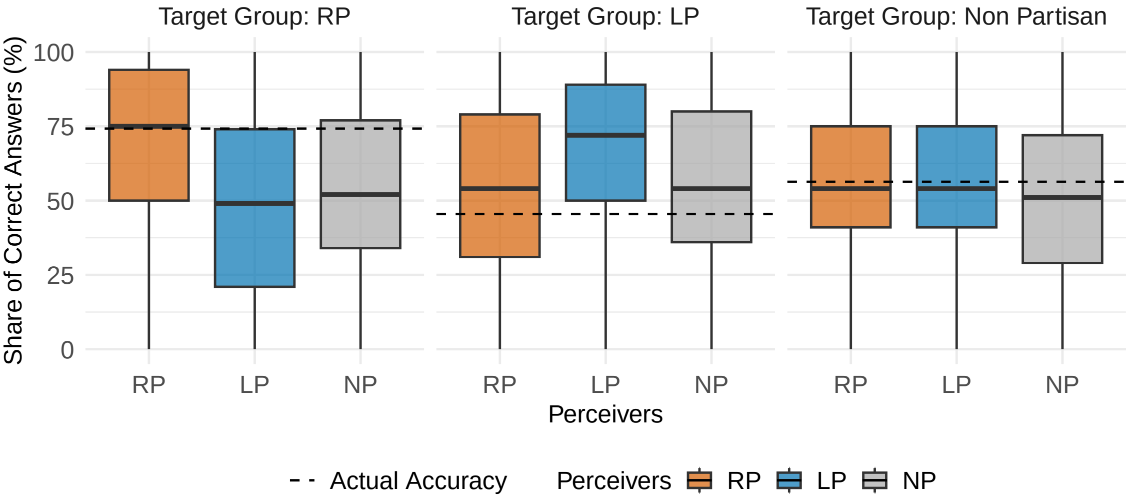
Fact 8 : The highest mountain in the United States is Denali (formerly known as Mt. McKir
Target Group: RP
Target Group: LP
Target Group: Non Partisan
Share of Correct Answers (%)
100
75
50
25
0
RP
LP
NP
RP
LP
NP
RP
LP
NP
Perceivers
Actual Accuracy
Perceivers
RP
LP
NP

Fact 9 : The Democratic Party stole the 2020 presidential election.




Fact 10 : Climate change is a hoax created to push socialist policies and destroy America
Target Group: RP
Target Group: LP
Target Group: Non Partisan
Share of Correct Answers (%)
100
75
50
25
0
RP
LP
NP
Perceivers
Actual Accuracy
Perceivers
RP
LP
NP

Fact 11 : The Republican administration initiated the Iraq war for oil interests.



Fact 12 : The Republicans stole the 2024 presidential election.